```
library(TwoSampleMR)
 library(tidyverse)
 library(data.table)
 library(dplyr)



protein_map <- fread(
 "/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/data/olink_protein_map_3k_v1
(1).tsv"
 )



protein_name <- "IL2"
 gene_info <- protein_map %>%
   filter(HGNC.symbol == protein_name)



cis_chr   <- gene_info$chr
 cis_start <- gene_info$gene_start - 1e6
 cis_end   <- gene_info$gene_end + 1e6





untar("/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/proteins_exposure_path/IL2/I
L2_P60568_OID20419_v1_Inflammation.tar",
      exdir = "IL2_data")





files <- list.files("IL2_data", full.names = TRUE, recursive = TRUE)



data <- rbindlist(lapply(files, fread))
```

```
IL2 <- data %>%
  mutate(
   LOG10P = as.numeric(LOG10P),
   pval = 10^(-LOG10P),
   ALLELE0 = tolower(ALLELE0),
   ALLELE1 = tolower(ALLELE1)
  ) %>%
  filter(
   CHROM == cis_chr,
   GENPOS >= cis_start,
   GENPOS <= cis_end
  )




setDT(IL2)
 map_path <- paste0(

"/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/rsid_maps/olink_rsid_map_mac5_i
nfo03_b0_7_chr",
 cis_chr,
 "_patched_v2.tsv.gz"
 )



rsid_map <- fread(map_path)
 setDT(rsid_map)



IL2[, GENPOS := as.integer(GENPOS)]



setkey(rsid_map, POS38)
 setkey(IL2, GENPOS)
 IL2 <- rsid_map[IL2]
 IL2[, SNP := rsid]
 IL2[, GENPOS := POS19]
 IL2 <- IL2[!is.na(SNP)]
```

```
clean_exposure_path <-
"/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/IL2_clean.txt"



fwrite(IL2, clean_exposure_path, sep="\t")
 IL2_Exp_dat <- read_exposure_data(
   filename = clean_exposure_path,
   sep = "\t",
   snp_col = "SNP",
   beta_col = "BETA",
   se_col = "SE",
   effect_allele_col = "ALLELE1",
   other_allele_col = "ALLELE0",
   eaf_col = "A1FREQ",
   pval_col = "pval",
   samplesize_col = "N"
 )



IL2_Exp_dat$exposure <- protein_name



IL2_Exp_dat <- clump_data(
 IL2_Exp_dat,
 clump_kb = 1000,
 clump_r2 = 0.01,
 pop = "EUR" )



outcome_path <- "/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/regenie
files/traitI518_merged.regenie"

 I518 <- fread(outcome_path) %>%
   mutate(
     LOG10P = as.numeric(LOG10P),
     pval = 10^(-LOG10P),
     ALLELE0 = tolower(ALLELE0),
     ALLELE1 = tolower(ALLELE1),
     SNP = ID
   )
```

```
clean_outcome_path <-
"/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/I518_clean.txt"
 fwrite(I518, clean_outcome_path, sep="\t")



I518_Out_dat <- read_outcome_data(
  filename = clean_outcome_path,
  sep = "\t",
  snp_col = "SNP",
  beta_col = "BETA",
  se_col = "SE",
  effect_allele_col = "ALLELE1",
  other_allele_col = "ALLELE0",
  eaf_col = "A1FREQ",
  pval_col = "pval",
  samplesize_col = "N"
 )



dat <- harmonise_data(
  exposure_dat = IL2_Exp_dat,
  outcome_dat  = I518_Out_dat
 )





mr_results_IL2 <- mr(
  dat,
  method_list = c(
    "mr_ivw",
    "mr_weighted_mode",
    "mr_weighted_median",
    "mr_egger_regression"
  )
 )
```

```
print(mr_results_IL2)
 write.csv(mr_results_IL2,
        file =
"/exports/eddie/scratch/v1ielli3/ComplexCVDTraits/results/mr_results_IL2.csv",
        row.names = FALSE)
```